

Hospital Operations & Patient Analytics Dashboard

Project Documentation

Submitted as part of

Infosys Springboard Virtual Internship 7.0 – Batch 2

PROJECT DETAILS

Project Information

Particulars	Details
Project Title	Hospital Operations & Patient Analytics Dashboard
Project Start Date	28 July 2026
Project End Date	17 September 2026
Internship Program	Infosys Springboard Virtual Internship 7.0 – Batch 2
Mentor Name	Prathina Likhita
Project Domain	Healthcare Data Analytics
Data Source	Kaggle
Visualization Tools	Microsoft Power BI and Tableau
Programming Language	Python

Project Description

The project focuses on analyzing hospital operations and patient-related data through data processing, KPI calculation, and interactive dashboard development.

The collected datasets are transformed into a structured format and used to develop dashboards that provide insights into hospital performance, patient admissions, department analytics, and resource utilization.

Project Deliverables

- Processed hospital datasets.
- Python scripts and Jupyter Notebook.
- Hospital KPI dataset in Excel format.
- Interactive Power BI dashboards.
- Interactive Tableau dashboards.
- Integrated dashboard files and project documentation.

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PROJECT OVERVIEW

Introduction

MedTrack_DV is a hospital operations and patient analytics project developed to understand hospital performance through data analysis and visualization.

The project uses publicly available hospital datasets collected from Kaggle. Python is used to process and prepare the data, while Microsoft Excel is used to store the final analytical dataset.

Power BI and Tableau are used to create interactive dashboards that present hospital-related information in a clear and understandable manner.

The project covers patient admissions, hospital performance, department analytics, and healthcare resource utilization.

By transforming raw data into meaningful visual insights, the project supports the understanding of hospital operations and helps identify areas that may require improvement.

Project Scope

The project includes:

- Collection and integration of hospital datasets.
 - Data cleaning and transformation.
 - Calculation of hospital performance indicators.
 - Development of interactive dashboards using Power BI and Tableau.
 - Integration and validation of dashboard files.
-

PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

Hospitals generate large amounts of patient and operational data. Understanding this information can be difficult when the data is stored in separate files and requires manual analysis.

Without a structured analytical approach, identifying patient admission patterns, comparing department performance, and monitoring resource utilization can become challenging.

This project addresses these challenges by processing hospital datasets and presenting the results through interactive dashboards.

Objectives

1. Collect hospital-related datasets from Kaggle.
2. Clean and transform the collected data using Python.
3. Calculate important hospital performance indicators.
4. Analyze patient admissions and hospital operational trends.
5. Develop interactive dashboards using Power BI and Tableau.
6. Present hospital data through meaningful visualizations.
7. Support data-driven understanding of hospital operations and resource utilization.

Project Scope

The analysis focuses on the available dataset fields and the hospital performance indicators developed during the project.

The dashboards are intended for analytical and educational purposes and do not replace clinical decision-making.

TOOLS AND TECHNOLOGIES

Technologies Used

Tool	Purpose
Kaggle	Collecting publicly available hospital datasets
Python	Data processing, cleaning, and KPI generation
Visual Studio Code	Writing and executing Python scripts
Jupyter Notebook	Performing and documenting data cleaning
Pandas and NumPy	Data manipulation and transformation
Microsoft Excel	Storing the final analytical dataset
Power BI	Developing interactive dashboards
Tableau	Creating additional visualizations and dashboards

Technology Overview

Kaggle serves as the source of the raw hospital data. Python is used to transform the collected files into a structured dataset suitable for analysis.

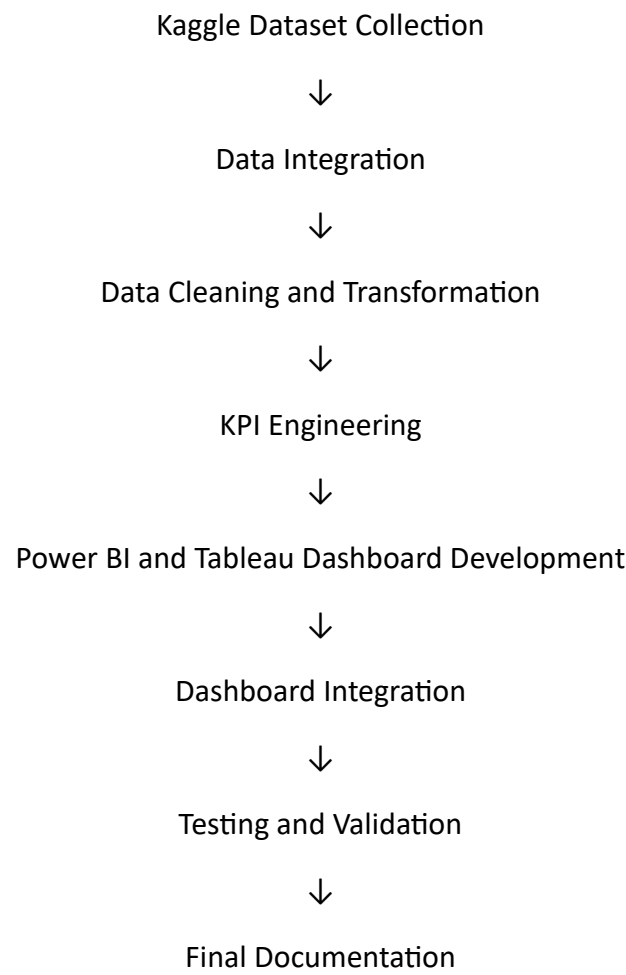
Jupyter Notebook provides an environment for performing data cleaning, while Visual Studio Code is used to develop and execute Python scripts.

The processed dataset is stored in Excel format and used as the basis for dashboard development in Power BI and Tableau.

The two visualization tools provide different ways to explore the same hospital analytical data.

PROJECT WORKFLOW AND METHODOLOGY

Workflow



Module-Wise Workflow

Module	Main Activity
1	Collect and integrate raw datasets
2	Clean and transform hospital data
3	Generate KPIs and final dataset
4	Develop initial Power BI dashboards
5	Develop Tableau dashboards and reuse selected Power BI dashboards
6	Develop additional department and resource dashboards
7	Merge and organize the dashboard files

The workflow ensures that the dashboards are developed using the processed analytical dataset rather than directly relying on unprocessed raw files.

MODULE 1: DATA COLLECTION

Module 1 – Hospital Data Collection

Objective

To collect the required hospital datasets from Kaggle and combine them into a single raw dataset for further analysis.

Dataset Collection

File Name	Description
admission.csv	Hospital admission records
bed.csv	Hospital bed-related data
patient.csv	Patient-related data
ward.csv	Hospital ward-related data

Procedure

1. Identify and download the required datasets from Kaggle.
2. Organize the downloaded CSV files.
3. Use Python code to process and integrate the datasets according to the project procedure.
4. Combine the required information into a single raw dataset.
5. Save the integrated output as Hospital_raw_dataset.csv.

Output

Hospital_raw_dataset.csv

Result

The separate hospital datasets are combined into one raw dataset, providing a common input for the data cleaning process in Module 2.

MODULE 2: DATA CLEANING AND TRANSFORMATION

Module 2 – Data Preparation

Objective

To prepare the raw hospital dataset for reliable analysis and visualization.

Data Cleaning Process

The raw dataset is processed using Python code and documented in a Jupyter Notebook. The cleaning process includes:

- Removing duplicate records.
- Handling missing patient data.
- Standardizing department names.
- Normalizing healthcare indicators.
- Preparing a structured dataset for KPI calculation.

Procedure

1. Load the raw dataset generated in Module 1.
2. Examine the dataset for inconsistencies.
3. Apply the required cleaning and transformation procedures.
4. Prepare the cleaned dataset for further analysis.
5. Save the processed data as `hospital_cleaned.csv`.
6. Document the cleaning procedure in `hospital_cleaning.ipynb`.

Output Files

- `hospital_cleaned.csv`
- `hospital_cleaning.ipynb`

Result

The raw hospital data is transformed into a cleaner and more consistent dataset, which serves as the input for KPI engineering in Module 3.

MODULE 3: KPI ENGINEERING

Module 3 – Hospital KPI Development

Objective

To calculate hospital performance indicators and prepare the final analytical dataset for dashboard development.

Input

hospital_cleaned.csv

Procedure

1. Load the cleaned hospital dataset.
2. Use Python code to calculate the required hospital performance indicators.
3. Organize the calculated results into a structured dataset.
4. Generate the final analytical dataset in Excel format.
5. Prepare the output for use in Power BI and Tableau.

Output Files

- generate_hospital_kpis.py
- hospital_final_dataset.xlsx

Result

The cleaned hospital data is transformed into a final analytical dataset containing the calculated hospital performance indicators.

This dataset is used as the foundation for the dashboard development modules.

HOSPITAL PERFORMANCE INDICATORS

Key Performance Indicators

Key Performance Indicators (KPIs) are used to summarize hospital operations and provide measurable information about performance.

The project includes the following hospital performance indicators.

KPI	Purpose
Total Admissions	Measures the number of hospital admissions
Occupancy Rate	Indicates the extent of hospital occupancy
Average Length of Stay	Measures the average duration of patient hospitalization
Readmission Rate	Indicates the frequency of patient readmissions
Bed Utilization Rate	Measures the utilization of available hospital beds
Department Efficiency Score	Supports comparison of department performance

Importance of KPIs

These indicators provide a summarized view of hospital operations and help users identify trends and differences in performance.

The calculated KPIs are incorporated into the dashboards to make hospital data easier to understand.

MODULE 4: POWER BI DASHBOARD DEVELOPMENT

Module 4 – Initial Power BI Dashboards

Objective

To develop interactive dashboards using the final hospital analytical dataset.

Input

hospital_final_dataset.xlsx

Dashboard Development

Power BI is used to transform the final analytical dataset into visual representations of hospital operations.

The dashboards use visualizations, KPI displays, filters, and navigation features to make the analysis easier to explore.

Development Procedure

1. Import the final dataset into Power BI.
2. Prepare the required fields for visualization.
3. Create the initial dashboard s.
4. Add charts and KPI visualizations.
5. Configure filters and dashboard navigation.
6. Review the dashboard layout and functionality.

Result

The initial Power BI dashboards provide a visual foundation for hospital and patient analytics.

Hospital Overview

department_name

All

admission_type

Emergency

admission_status

All

gender

Female

admission_date

All

Hospital Overview

Patient Flow

Department Analytics

Resource Utilization

45K

Total Admissions

55.37K

Occupancy Rate

6

Average LOS

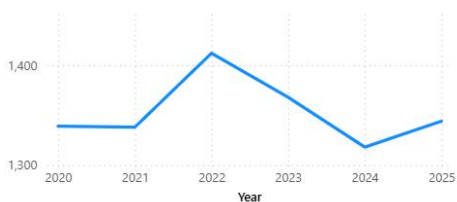
6

Readmission Rate

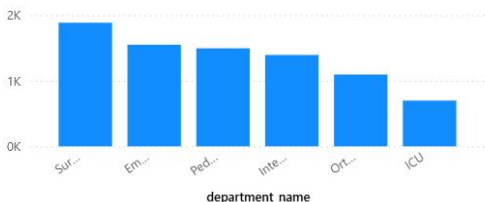
53

Bed Utilization Rate

Count of admission_id by Year



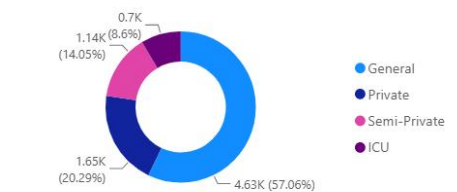
Count of admission_id by department_name



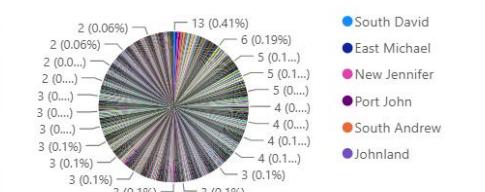
Sum of Occupancy Rate (%)



Count of admission_id by ward_type



Count of admission_id by city



Patient Flow

gender

All

blood_group

All

admission_status

All

city

Aaronside

Unique Patients

All

Hospital Overview

Patient Flow

Department Analytics

Resource Utilization

45K

Total Admissions

8

Total LOS

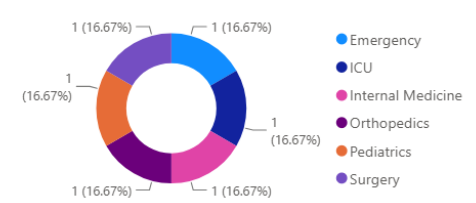
2

Total readmitted

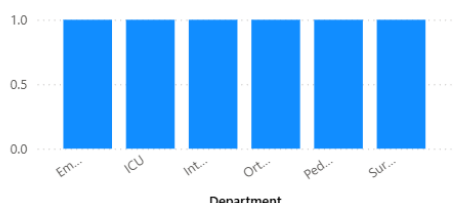
30K

Total Patients

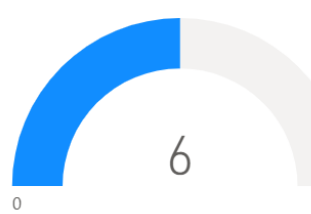
Count of Unique Patients by Department



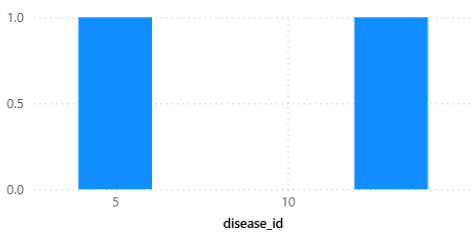
Count of Average LOS (Days) by Department



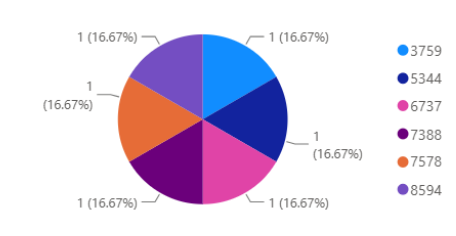
Count of Readmission Rate (%)



Count of gender by disease_id



Count of Readmission Rate (%) by Unique Patients



Patient Flow

Department Analytics

Department

Emergency

Available Beds

All

Bed Utilization Rate (%)

All

ward_name

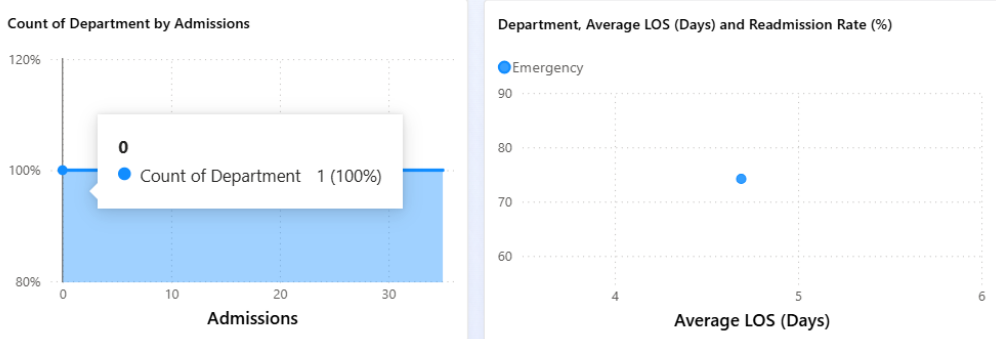
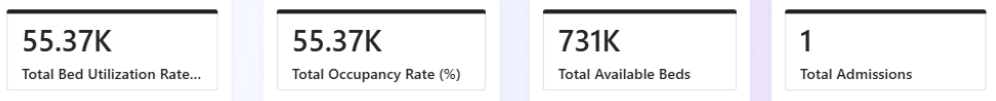
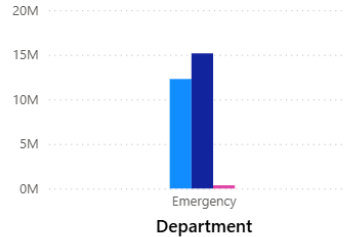
Emergency Ward 2

ward_type

Private

Sum of patient_id by Department and gender

Female Male Other



Resource utilization

Total Beds

All

city

Aaronborough

contact_number

All

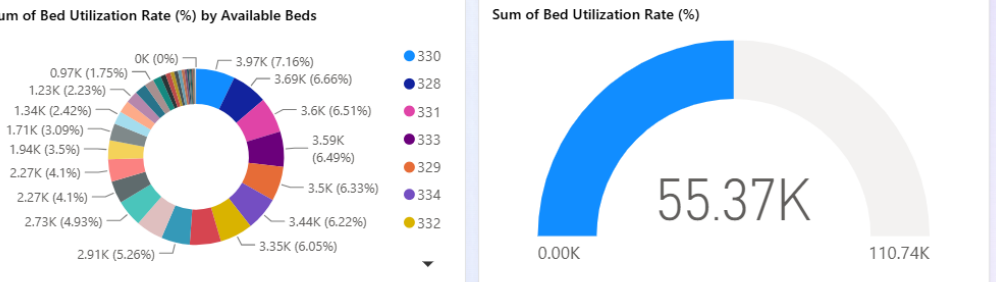
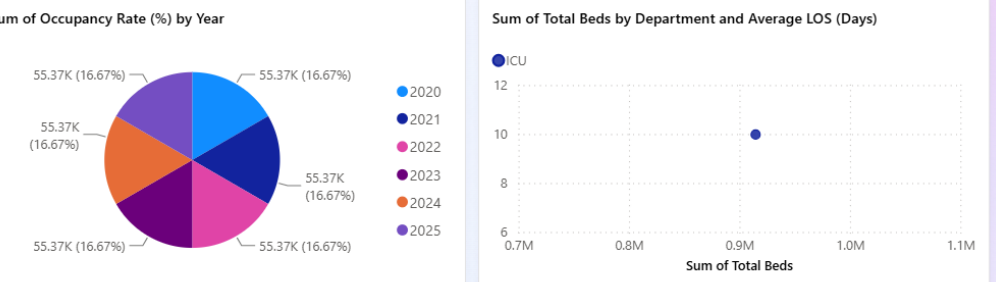
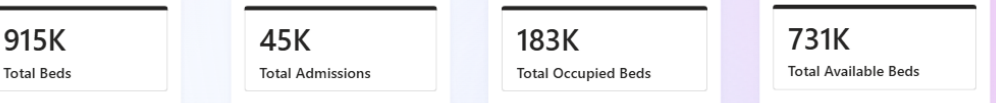
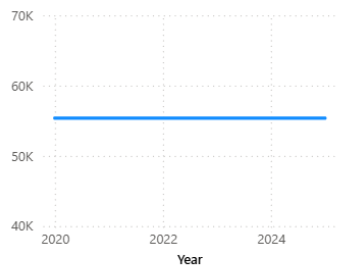
ward_type

All

Department

ICU

Sum of Bed Utilization Rate (%) by Year



POWER BI DASHBOARD ANALYSIS

Power BI Dashboard Overview

The Power BI dashboards present hospital information through interactive visualizations.

Dashboard 1 – Hospital Overview

This dashboard provides an overall view of hospital performance and presents the important performance indicators.

It helps users understand hospital admissions, occupancy, and operational trends.

Dashboard 2 – Patient Analysis

This dashboard focuses on patient-related information and admission patterns.

It supports the exploration of patient distribution and other available patient-related indicators.

Dashboard Interaction

The dashboards use filters and navigation features to allow users to explore different sections of the analysis.

The interactive design helps users examine hospital information without manually searching through the raw dataset.

MODULE 5: TABLEAU DASHBOARD DEVELOPMENT

Module 5 – Tableau Visualization

Objective

To extend the dashboard analysis using Tableau while retaining the first two Power BI dashboards.

Development Approach

The first two Power BI dashboards are retained as part of the overall project.

Two additional dashboards are developed using Tableau to provide another visualization environment for hospital analytics.

Procedure

1. Use the final analytical dataset generated in Module 3.
2. Connect the dataset to Tableau.
3. Prepare the required fields for visualization.
4. Develop the first two Tableau dashboards.
5. Configure the required filters and navigation.
6. Review the dashboards for consistency with the project objectives.

Result

The project combines the initial Power BI dashboards with additional Tableau dashboards, expanding the overall dashboard suite.

Hospital Overview

Hospital Overview	Patient Flow	Department Analytics	Resource Utilization	
Total admissions	Average LOS	Bed Utilization Rate	Occupancy Rate	Readmission Rate
10,126	5	25	25	73

Filters

Gender
Female

Admission Status
(All)

Admission Type
Elective

Department Name
(All)

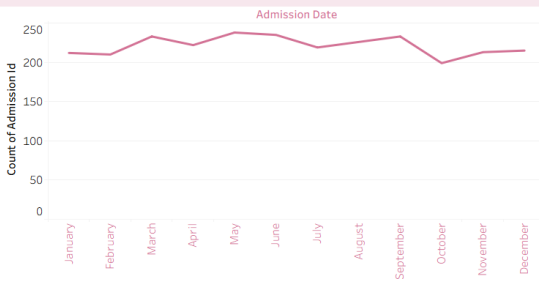
Ward Type Distribution



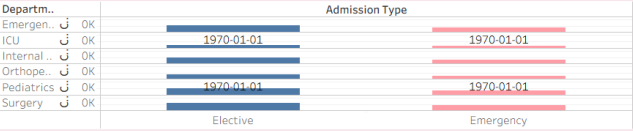
Patient Distribution by Blood Group



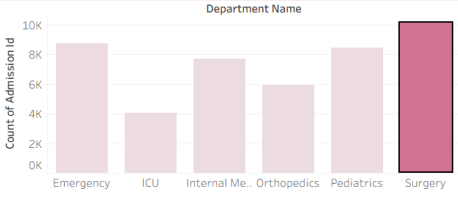
Monthly Admission trends



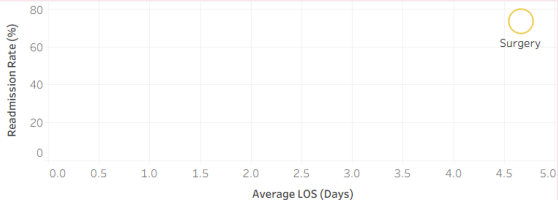
Admission Type by Department



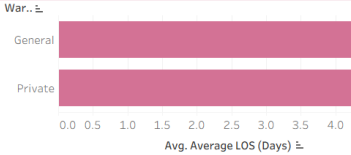
Average Length of Stay by Ward Type



Department-wise Average Length of Stay vs Readmission Rate



Average Length of Stay by Ward Type



Patient Analysis

Hospital Overview	Patient Flow	Department Analytics	Resource Utilization
Total LOS	Total Patients	Total Readmission	Total admissions
231,975	39,400	449	45,000

Filters

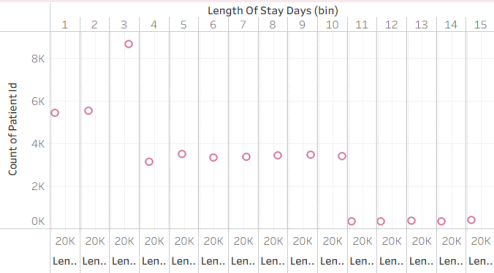
City
Aaronberg

Admission Status
(All)

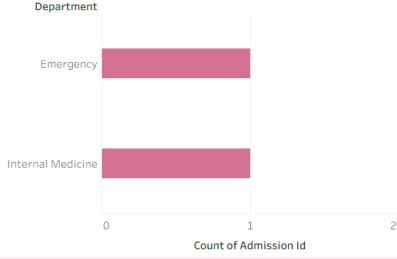
Blood Group
A-

Gender
Female

Length of Stay Distribution



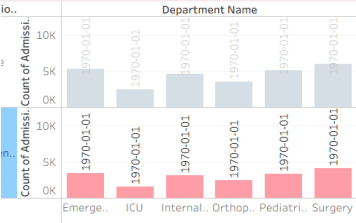
Department-wise Admissions



Patient Distribution by Blood Group



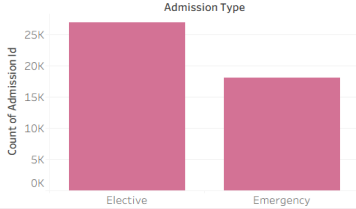
Admission Type by Department



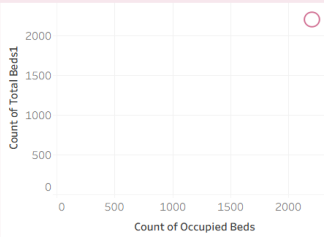
Patient Distribution by Department



Admissions by Admission Type



Bed Distribution Analysis



MODULE 6: DEPARTMENT AND RESOURCE ANALYTICS

Module 6 – Additional Tableau Dashboards

Objective

To complete the remaining dashboard analysis using the existing Power BI dashboards and additional Tableau dashboards.

Power BI Dashboards

Power BI dashboards 3 and 4, developed during Module 4, are retained in the project.

Tableau Dashboards

Tableau dashboards 3 and 4 are developed to provide additional visual analysis of hospital operations.

Analysis Areas

The project plan identifies department analytics and resource utilization as important areas of hospital analysis.

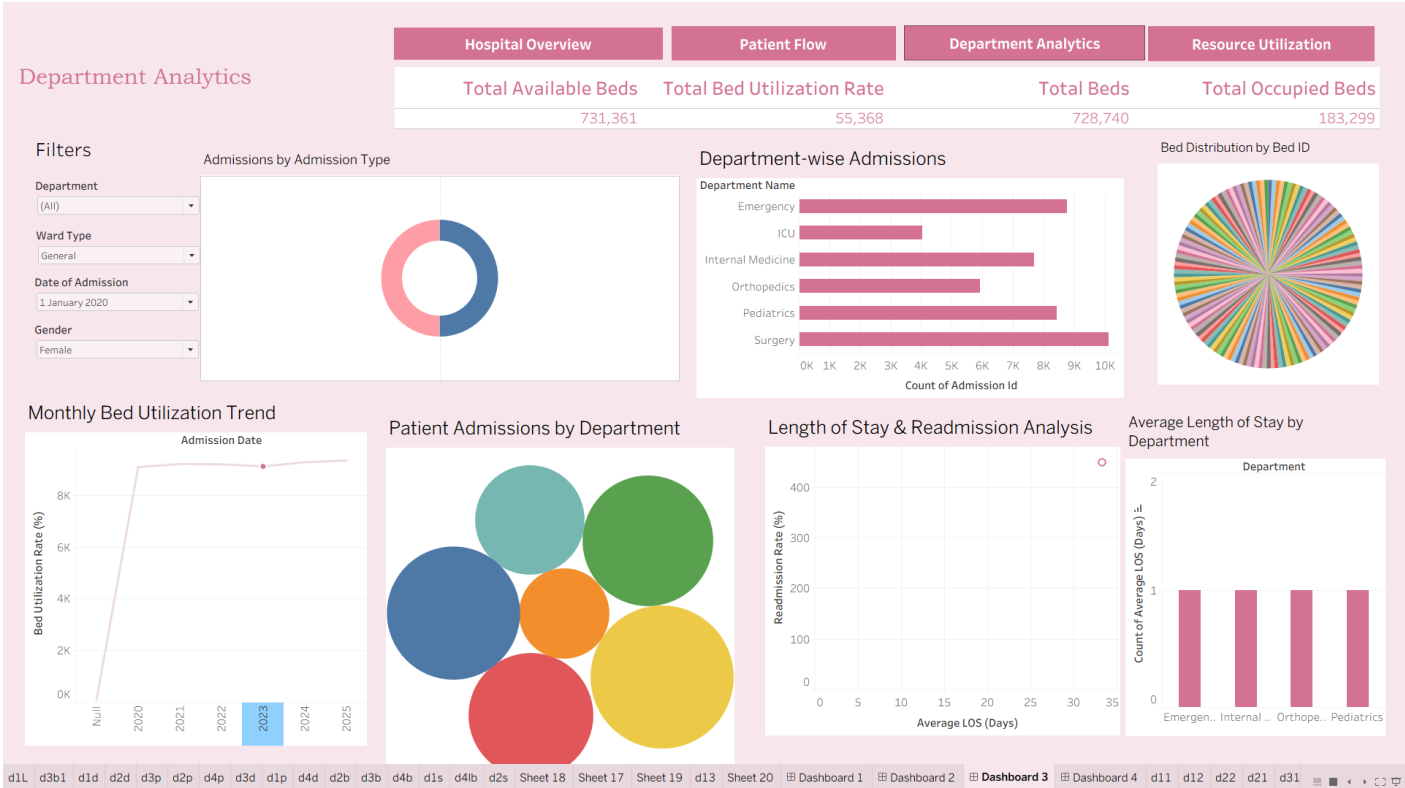
These areas include:

- Department performance analysis.
- Patient volume by department.
- Department efficiency comparison.
- Bed utilization analysis.
- Resource availability analysis.
- Capacity planning insights.

The exact visualizations and metrics depend on the fields available in the final analytical dataset.

Result

The remaining dashboards complete the visual analysis of the project and provide additional views of hospital operations and resource utilization.



MODULE 7: DASHBOARD INTEGRATION

Module 7 – Final Dashboard Integration

Objective

To organize and combine the developed dashboards into the final project deliverables.

Integration Process

The dashboards developed during the previous modules are organized into separate Power BI and Tableau dashboard files.

The integration process includes:

1. Collect the completed Power BI dashboards.
2. Collect the completed Tableau dashboards.
3. Organize the dashboard files.
4. Review the dashboards for consistency.
5. Prepare the final dashboard deliverables.

Final Deliverables

- Power BI dashboard file.
- Tableau dashboard file.

Result

The dashboard development work is organized into final Power BI and Tableau deliverables, making the project easier to present, review, and maintain.

TESTING AND VALIDATION

Dashboard Testing and Validation

Objective

To verify that the processed datasets, KPI calculations, and dashboard interactions function as expected.

Testing Activities

Testing Area	Purpose
Dataset Validation	Check the processed dataset used for analysis
KPI Validation	Review the calculated hospital indicators
Visualization Testing	Check whether charts display the intended data
Filter Testing	Verify that filters support the required analysis
Navigation Testing	Check dashboard navigation and transitions
Integration Testing	Review the organized Power BI and Tableau deliverables

Validation Procedure

1. Review the final analytical dataset.
2. Check the KPI values against the generated results.
3. Examine the dashboard visualizations.
4. Test the available filters and navigation features.
5. Review the final dashboard files.

Result

Testing and validation help identify inconsistencies and ensure that the dashboards present the processed hospital data in an understandable and consistent manner.

PROJECT OUTCOMES AND APPLICATIONS

Project Outcomes

The project demonstrates the use of data analytics and visualization techniques to understand hospital operations.

Major Outcomes

- Integrated multiple hospital datasets into a single raw dataset.
- Prepared a cleaned dataset using Python.
- Generated hospital performance indicators.
- Created a final analytical dataset in Excel format.
- Developed dashboards using Power BI and Tableau.
- Organized the dashboards into final project deliverables.

Applications

The dashboard suite can support the understanding of:

- Hospital admission patterns.
- Patient-related trends.
- Department performance.
- Hospital occupancy.
- Bed and resource utilization.

Project Limitations

The analysis is dependent on the availability, completeness, and structure of the Kaggle datasets.

The dashboards represent the information contained in the prepared analytical dataset and should not be interpreted as real-time hospital monitoring unless connected to an appropriate live data source.

The project is intended to support data analysis and educational purposes.

CONCLUSION AND REFERENCES

Conclusion

MedTrack_DV – Hospital Operations & Patient Analytics Dashboard demonstrates how hospital data can be transformed into meaningful visual insights through a structured data analytics workflow. The project begins with the collection of hospital datasets from Kaggle and continues through data integration, cleaning, transformation, and KPI engineering using Python. The final analytical dataset is then used to develop interactive dashboards in Power BI and Tableau. Through these dashboards, the project presents hospital performance, patient-related information, department analytics, and resource utilization in a visual and accessible format. The project provides practical experience in data processing, KPI development, dashboard design, and healthcare data analytics.

References

1. Kaggle – Source of the hospital datasets used in the project.
2. HOPA.pdf – Project plan and module-wise implementation procedure.
3. Python documentation – Reference for Python-based data processing.
4. Pandas documentation – Reference for data manipulation and cleaning.
5. Microsoft Power BI – Dashboard development tool.
6. Tableau – Data visualization and dashboard development tool.

Final Project Deliverables

- Hospital_raw_dataset.csv
- hospital_cleaned.csv
- hospital_cleaning.ipynb
- generate_hospital_kpis.py
- hospital_final_dataset.xlsx
- Final Power BI dashboard file
- Final Tableau dashboard file